Data Science & Artificial Intelligence

Data ware housing

Data Cleaning & Smoothing

DPP-02

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[MCQ]

#Q. Which of the following is not a data cleaning technique?

A. Outlier detection and removal

B. Imputation of missing values

C. Data transformation

D. Data segmentation

Related to data partioning rather than cleaning

[MCQ]

#Q. In data cleaning, what is outlier detection used for?

A. To add noise to the data

B. To remove incorrect or irrelevant data points

C. To balance the data distribution

D. To normalize the data range

[MCQ]

#Q. Which of the following is an example of data smoothing technique?

A. Principal Component Analysis (PCA)

B. Moving Average — Reduce noise

averaging neighbour data point

C. Logistic Regression

D. K-means clustering

[MCQ]

#Q. What does the term "data deduplication" refer to in data cleaning?

A. Adding duplicate entries to the dataset

B. Detecting and removing duplicate entries from the dataset

C. Splitting the dataset into smaller segments

D. Converting categorical data to numerical data

[MCQ]

#Q. Which of the following is a disadvantage of data smoothing techniques?

A. They reduce the interpretability of data patterns

B. They increase the likelihood of overfitting

C. They can only handle numerical data

D. They are computationally expensive

[MCQ]

#Q. What is the primary goal of data normalization?

A. To reduce the dimensionality of the dataset

B. To standardize the range of numerical data

C. To introduce noise into the dataset

D. To handle missing values in the dataset

Transform the data into standard Range

[MCQ]

#Q. Which of the following normalization techniques is robust against outliers?

A. Min-Max scaling

B. Z-score normalization – less affected by outlier Min Max

C. Log transformation

D. Unit vector scaling

[MCQ]

#Q. When should you use logarithmic transformation for data normalization?

A. When the data has a skewed distribution

B. When the data has a normal distribution

C. When the data has missing values

D. When the data is categorical

[MCQ]

#Q. Which normalization technique is suitable for sparse datasets with mostly zero values?

A. Min-Max scaling

B. Z-score normalization

C. Log transformation

D. Unit vector scaling — normalizes each sample (row) in to dataset.

[MCQ]

#Q. What does Min-Max scaling do to the data?

A. Centers the data around zero with unit variance

B. Standardizes the range of data to a fixed range (e.g., [0, 1])

C. Transforms the data to a logarithmic scale

D. Normalizes the data to have unit norm

[MCQ]

#Q. What is data generalization in the context of data mining?

A. Reducing the volume of data for efficient storage

B. Generating summary or aggregated data from detailed data

C. Converting continuous data into categorical data

D. Normalizing the range of numerical data

[MCQ]

#Q. Which of the following is an example of data aggregation?

Combining data sets

A. Grouping sales transactions by product category

B. Converting temperature readings from Celsius to Fahrenheit

C. Removing outliers from a dataset

D. Scaling numerical data to a fixed range

[MCQ]

#Q. Which of the following is NOT a step in data cleaning?

A. Handling missing values

B. Removing duplicates

C. Aggregating data

D. Correcting inconsistent data

[MCQ]

#Q. Which data smoothing technique is used to replace missing values with the most recent non-missing value?

A. Moving average

B. Exponential smoothing

C. Last observation carried forward (LOCF)

D. Savitzky-Golay filter

[MCQ]

#Q. Which of the following normalization techniques can handle outliers better?

A. Min-Max normalization

B. Z-score normalization

C. Decimal scaling

D. Log transformation

THANK - YOU